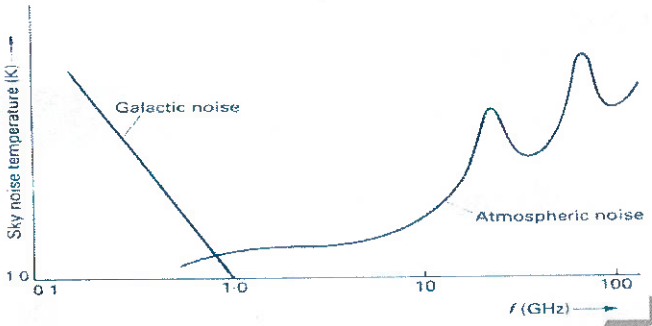
Model answer to Q1(a): Bookwork

Describe the two main contributions to the clear sky noise temperature between 0.1 GHz and 10 GHz.



[3]

Model answer to Q1(b): Bookwork

From 1(a), what is the approximate frequency range that has minimum atmospheric noise. State the ubiquitous application found at such frequencies.

Minimum atmospheric noise occurs between 0.8 to 1 GHz with mobile phone applications found over this frequency range.

[2]

Model answer to Q1(c): Calculation

Calculate the approximate wavelength and wavenumber in free space for 1(b).

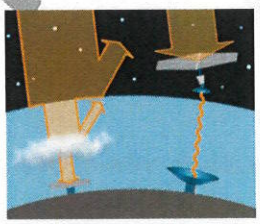
At approximately 0.9 GHz, the wavelength is c/f = 33.3 cm and wavenumber is 3 per metre.

[1]

Model answer to Q1(d): Bookwork

Explain the principle behind the microwave-based Space Solar Power System (M-SSPS) for wireless energy transmission.

With conventional terrestrial-based solar farms, part of the solar energy is lost on its way through the atmosphere by the effects of reflection and absorption.



The Japan Aerospace Exploration Agency (JAXA) is proposing a Microwave-based Space Solar Power System (M-SSPS) for wireless energy transmission, whereby a satellite having large solar panel arrays generates dc power and most of this energy is transmitted to earth via a microwave link at 5.8 GHz (which exhibits relatively low atmospheric attenuation). The geostationary satellite is positioned 36,000 km directly above the earth's 3 km diameter circular receiving target area. The 5.8 GHz Wireless Power Transfer technology has ~12% efficiency.

[4]

Model answer to Q1(e): Bookwork

Explain the engineering compromises associated with the proposed frequency of operation for 1(d).

The frequency cannot be too low, otherwise the on-board satellite transmit antenna would be too large to produce a pencil beam that would have a footprint not extending beyond the size of the

Answers

terrestrial-based rectenna array. If the frequency is too large then both the on-board satellite transmitter and terrestrial-based rectenna array would be too inefficient in terms of transmission and reflection losses.

[2]

Model answer to Q1(f): Bookwork

Write down Ampere's law of magnetomotive force.

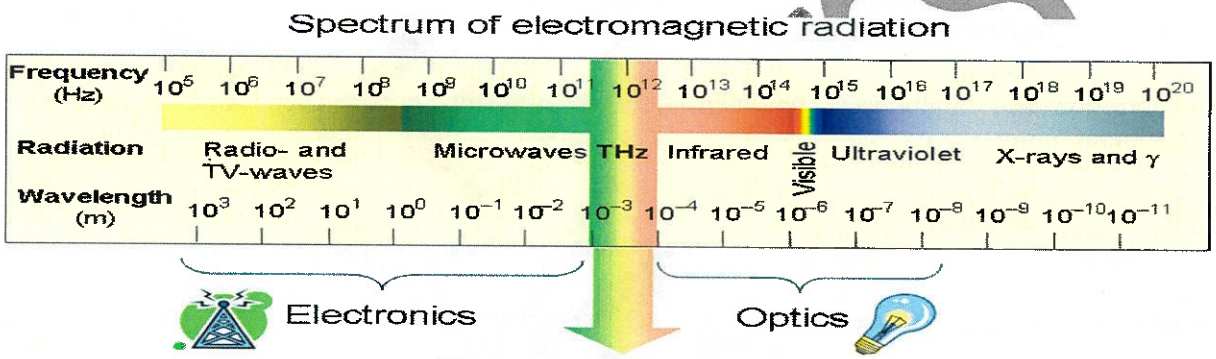
$(0.1 \Rightarrow 0.3)$ THz Gap $(3 \Leftarrow 10)$ Transition from Electronics to Optics

Ampere's Law of Magnetomotive Force: $\nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}$

[2]

Model answer to Q1(g): Bookwork

From 1(f), which term is associated with optics/photonics? At which approximate wavelength and frequency does the performance of optics/photonics begin to drop off?



Optical/photonic systems drop off in performance at wavelengths longer than 100 microns or frequencies less than 3 THz.

[2]

Model answer to Q1(h): Bookwork

From 1(f), which term is associated with electronics? At which approximate wavelength and frequency does the performance of electronics begin to drop off?

Electronic systems drop off in performance at wavelengths shorter than 1 mm or frequencies greater than 0.3 THz.

[2]

Model answer to Q1(i): Bookwork

Briefly comment on the 'THz Gap' and how it is narrowing over time in frequency.

The 'THz gap' is associated with the frequency range in which it is difficult to generate electromagnetic radiation; with electronic systems performance dropping off above 0.3 THz and photonic systems falling off at frequencies below 3 THz. In the past, these respective values were 0.1 THz and 10 THz and so the gap is closing.

[2]

Answers

Model answer to Q2(a): New Derivation

Using the over-simplified linear strip approach, derive the total inductance per unit length in terms of classical skin depth. In addition, derive the parallel-plate capacitance per unit length for the average circumference of the insulator and the associated characteristic impedance.

$$L' = \frac{L_{inner}}{l} + \frac{L_{outer}}{l} \sim \frac{\mu_o \delta_o / 2}{\pi d} + \frac{\mu_o \delta_o / 2}{\pi D} = \frac{\mu_o \delta_o}{2\pi}\left(\frac{1}{d} + \frac{1}{D}\right) [H/m]$$

where classical skin depth $\delta_o = \sqrt{\frac{2\rho_o}{\omega \mu_o}}$

$$C' = \frac{C}{l} \sim \varepsilon_o \varepsilon_r \frac{\pi\left(\frac{D+d}{2}\right)}{(D-d)} = \frac{\pi \varepsilon_o \varepsilon_r}{2} \frac{(D+d)}{(D-d)} [pF/m]$$

$$Zo = \sqrt{\frac{L'}{C'}} [\Omega]$$

[4]

Model answer to Q2(b): New Calculation

From 2(a), for the RG-6/U cable, calculate the classical skin depth at 25 GHz and associated inductance per unit length. In addition, calculate the capacitance per unit length and the associated characteristic impedance.

$$\delta_o = 0.4126 \text{ µm}$$

$$L' \sim (81.22 + 18.05)\, pH/m = 99.27\, pH/m$$

$$C' \sim 32.50\, pF/m$$

$$Zo \sim 1.75\ \Omega$$

[4]

Model answer to Q2(c): New Calculation

For the RG-6/U coaxial cable, calculate the inductance and capacitance per unit length and associated characteristic impedance.

$$L' = 300.8\, [nH/m]$$

$$C' = 55.0\, [pF/m]$$

$$Zo = \sqrt{\frac{L'}{C'}} = \frac{1}{2\pi}\sqrt{\frac{\mu}{\varepsilon}} ln\left\{\frac{D}{d}\right\} = \frac{\eta_o}{2\pi}\sqrt{\frac{\mu_r}{\varepsilon_r}} ln\left\{\frac{D}{d}\right\} \sim \frac{60}{\sqrt{\varepsilon_r}} ln\left\{\frac{D}{d}\right\} = 74.0\ \Omega$$

[2]

Model answer to Q2(d): Bookwork

From the calculated characteristic impedance in 2(c), what kind of application does this particular type of coaxial cable have and why?

A characteristic impedance of approximately 75 Ω gives the lowest attenuation for a coaxial cable. As a result, it is used to connect a TV to a remote antenna, where signal losses have to be kept to a minimum.

[2]

Answers

Model answer to Q2(e): New Conclusion

Comment on the frequency dispersive nature of the exact expressions, when compared to those derived in 2(a), and the calculated results for the RG-6/U cable. What conclusion can be made from the use of the over-simplified linear strip approach?

The exact expression is independent of frequency, as it is not a function of skin depth. However, the over-simplified linear strip approach is a function skin depth and frequency. The latter is 3,030 times lower than the exact value of inductance, while the capacitance is of a similar order. As a result, the over-simplified characteristic impedance is out by a factor of 42, making it useless.

[2]

Model answer to Q2(f): Discussion in Class

With a coaxial cable, what is the dominant mode of propagation and its associated lower cut-off frequency?

The dominant mode is the transverse electromagnetic (TEM) mode, having a lower cut-off frequency at DC

[2]

Model answer to Q2(g): New Derivation

The mode with the next lowest cut-off frequency is the TE11 mode. To a good approximation, the condition for the TE11 mode to propagate is that the wavelength in the dielectric is no longer than the average circumference of the insulator. Derive an expression for the cut-off frequency for the TE11 mode.

$$f_{TE11} \approx \frac{v}{\lambda_{TE11}} = \frac{c}{\pi\left(\frac{D+d}{2}\right)\sqrt{\varepsilon_r}}$$

[2]

Model answer to Q2(h): New Caculation

Using 2(g), for the RG-6/U coaxial cable, calculate the lower cut-off frequency for the TE11 mode and give a reasonable upper cut-off frequency for the dominant mode.

$$f_{TE11} \approx \frac{v}{\lambda_{TE11}} = \frac{\frac{c}{\sqrt{\varepsilon_r}}}{2\pi\left(\frac{R+r}{2}\right)} = \frac{c}{\pi\left(\frac{D+d}{2}\right)\sqrt{\varepsilon_r}} \rightarrow 28.03 \text{ GHz}$$

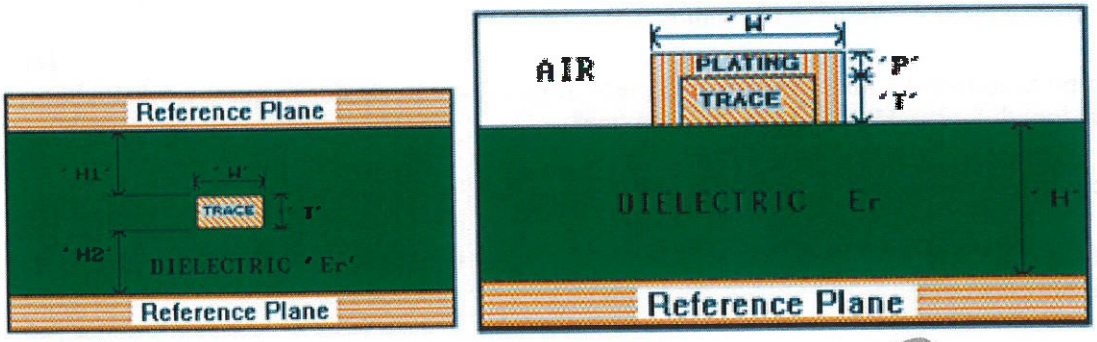
Safe to operate in the TEM mode up to ~90% of the lowest TE_{11} mode cut-off frequency, ~25 GHz.

[2]

Answers

Model answer to Q3(a): Discussion in Class

Sketch their cross-sections and comment on the relative frequencies of their transvers resonant modes.



Since the track width is wider with microstrip, for the same characteristic impedance, the transverse resonance frequency will be lower that that for stripline. However, since the effective relative permittivity for microstrip is less than that for microstrip (with non-air insulators) this may not always be true.

[2]

What is the difference in their dominant mode of propagation when the dielectric constant of the insulator is increased from the free-space value?

With a dielectric constant of unity the dominant mode of propagation is TEM for both types of transmission line. However, as the dielectric constant increases, only the stripline stays TEM, with microstrip becoming quasi-TEM.

[2]

State one prominent advantage for stripline and one for microstrip.

Stripline has inherent isolation from signals above and below the ground planes, while microstrip is easier to implement. Other advantages could also be stated.

[2]

Model answer to Q3(b): Discussion in Class

Beyond the distance L_1, does the track width increase or decrease? As a result, comment of the changes in the inductance per unit length and capacitance per unit length for this 35.355 Ω transmission line.

The track increases in width. As a result, the inductance per unit length decreases and the capacitance per unit length increases.

[2]

Calculate the incident voltage-wave reflection coefficient at the discontinuity. At the input port, how does the incident voltage-wave reflection coefficient change with frequency?

$$\rho = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{1 - \sqrt{2}}{1 + \sqrt{2}} = -0.1716$$

The voltage-wave reflection coefficient should not change with frequency.

[2]

Model answer to Q3(c): Discussion in Class

At the input port, what is the incident voltage-wave reflection coefficient at zero frequency?

At DC the transmission lines are effectively not seen and so the input port just sees the same terminating impedance. As a result, the reflection coefficient is zero.

[1]

Answers

At the input port, constructive interference will be seen at frequency f_1. At this frequency, what will be the electrical length for the 35.355 Ω transmission line?

The electrical length will be $\pi(0.5 + n)$, with $n = 0,1,2,3$ ...

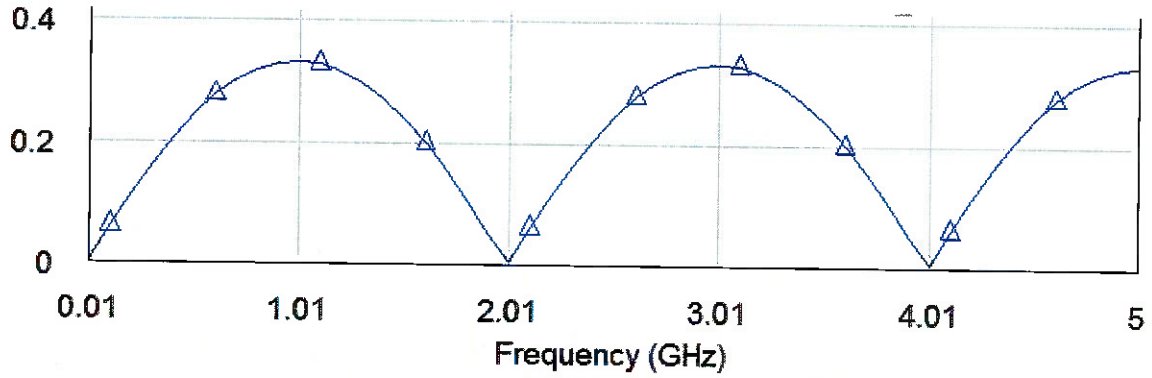
[2]

At the input port, destructive interference will be seen at frequency f_2. At this frequency, what will be the electrical length for the 35.355 Ω transmission line?

The electrical length will be $\pi(1 + n)$, with $n = 0,1,2,3$ ...

[2]

Sketch the frequency response for the magnitude of the incident voltage-wave reflection coefficient as seen at the input port.



[1]

Derive an expression for the difference frequency $\Delta f = (f_2 - f_1)$ in terms of the physical length of the 35.355 Ω transmission line L_2.

At f_1 the electrical length is 90 degrees. This corresponds to $L_2 = \lambda_g(f_1)/4$.
At f_2 the electrical length is 180 degrees. This corresponds to $L_2 = \lambda_g(f_2)/2$.

$$\lambda_g(f) = \frac{c/\sqrt{\varepsilon_r}}{f}$$

$$\Delta f = (f_2 - f_1) = \frac{c}{\sqrt{\varepsilon_r}}\left(\frac{1}{\lambda_g(f_2)} - \frac{1}{\lambda_g(f_2)}\right) = \frac{c/\sqrt{\varepsilon_r}}{4L_2}$$

[3]

With $\Delta f = 1$ GHz and the dielectric constant of the insulator being $\varepsilon_r = 2.45$, calculate L_2.

$$L_2 = \frac{c/\sqrt{\varepsilon_r}}{4\Delta f} = 48 \text{ mm}$$

[1]

Model answer to Q4(a): Textbook Detivation

Using the generic relationship for propagation constant γ as a starting point, given by equation (4.1), derive the expression for guided wavelength λ_g in terms of the free-space wavelength λ_o and cut-off wavelength λ_c.

$$\gamma_o^2 = \gamma_x^2 + \gamma_y^2 + \gamma_z^2 \qquad (4.1)$$

$$\gamma_g = \gamma_z = \sqrt{\gamma_o^2 - \gamma_x^2 + \gamma_y^2} \text{ where } \gamma_y = 0 \text{ for } TE_{10} \text{ mode}$$

Answ

$$\gamma_g = j\beta_g = j\frac{2\pi}{\lambda_g};\ \gamma_o = j\beta_o = j\frac{2\pi}{\lambda_o};\ \gamma_x = j\beta_c = j\frac{2\pi}{\lambda_c}$$

$$\beta_g^2 = \beta_o^2 - \beta_c^2 \text{ and } \left(\frac{2\pi}{\lambda_g}\right)^2 = \left(\frac{2\pi}{\lambda_o}\right)^2 - \left(\frac{2\pi}{\lambda_c}\right)^2$$

$$\lambda_g = \frac{\lambda_o}{\sqrt{1-\left(\frac{\lambda_o}{\lambda_c}\right)^2}}$$

[4]

Model answer to Q4(b): Textbook Derivation

What is the relationship between the cut-off wavelength for the dominant TE_{10} mode and the internal broad dimension a?

$$\lambda_c = 2a$$

[1]

Model answer to Q4(c): Discussion in Class

Using the expression derived in 4(a), what happens to the propagation constant below the cut-off frequency?

λ_g and β_g become purely imaginary and so γ_g becomes purely real, i.e. the wave attenuates as an evanescent mode without any propagation.

[2]

Model answer to Q4(d): New Derivation

Using the generic relationship between impedance η and propagation constant, given by equation (4.2), derive the wave impedance Z_{TE} in terms of the intrinsic impedance of free space η_o, free-space wavelength λ_o and cut-off wavelength λ_c.

$$\eta = \frac{j\omega\mu}{\gamma} \qquad (4.2)$$

$$Z_{TE} = \frac{j\omega\mu_o}{\gamma_g} = \frac{\omega\mu_o}{\beta_g} = \frac{\omega\mu_o}{\beta_o}\frac{1}{\sqrt{1-\left(\frac{\lambda_o}{\lambda_c}\right)^2}} = \frac{\eta_o}{\sqrt{1-\left(\frac{\lambda_o}{\lambda_c}\right)^2}}$$

[2]

Model answer to Q4(e): New Calculations

The WR-90 waveguide has internal broad dimensions of a = 22.86 mm. At an operating frequency $f_o = 1.5\ f_c$, where f_c is the cut-off frequency, calculate:

i) Cut-off frequency.

$$\lambda_c = 2a \text{ and } f_c = \frac{c}{\lambda_c} \cong 6.5617 \text{ GHz}$$

[1]

ii) Guided wavelength.

$$\frac{\lambda_o}{\lambda_c} = \frac{2}{3} \text{ and so } \lambda_g = \frac{\lambda_o}{\sqrt{1-\left(\frac{2}{3}\right)^2}} = 40.89 \text{ mm with } \lambda_o = c/f_o$$

[2]

iii) Physical length for the TE_{101} box mode.

$$L_z = \frac{\lambda_g}{2} = 20.45 \text{ mm}$$

[1]

iv) Wave impedance.

Answers

$$Z_{TE} = \frac{\eta_o}{\sqrt{1-\left(\frac{2}{3}\right)^2}} \approx 506\ \Omega \text{ with } \eta_o \approx 120\pi\ \Omega$$

[2]

Model answer to Q4(f): Discussion in Class

The slotted-line waveguide experiment can directly measure the magnitude of the standing wave $|V|$. Briefly explain the physical characteristics of the slot (e.g., its shape and position) and the diode detector's *I-V* characteristics that allow such measurements to be undertaken.

The non-radiating slot must be at the centre of the broad wall, where the E-field is at a maximum, and run along the direction of propagation (i.e., along the *z*-direction), so that lines of currents flow are not broken (which would otherwise result in radiation and, therefore, create associated losses). The Schottky diode detector normally has a square-law characteristic, which means that its output current is approximately proportional to the square of the E-field incident on the diode.

[2]

Model answer to Q4(g): New Application

The relationship between the magnitude of the voltage-wave reflection coefficient $|\rho|$ of the impedance terminating the end of the waveguide and slotted-line measured voltage standing wave ratio $VSWR = |V|_{MAX}/|V|_{MIN}$ is given by equation (4.3).

$$|\rho| = \frac{VSWR-1}{VSWR+1} \qquad (4.2)$$

Determine the *VSWR* when the waveguide is terminated by an open circuit at an operating frequency $f_o = 1.5\ f_c$, where f_c is the cut-off frequency.

$$VSWR = \frac{1+|\rho|}{1-|\rho|} = 1.3416$$

with $\rho = \frac{\eta_o - Z_{TE}}{\eta_o + Z_{TE}} = -0.1459$ and Return Loss = -16.7 dB

[3]

Model answer to Q5(a): Textbook

State the fundamental equation for quality factor in terms of energy.

$$Q(\omega) = 2\pi \frac{Total\ Reactive\ Energy\ Stored\ (W_L|_{PEAK}\ or\ W_C|_{PEAK})}{Average\ Resistive\ Energy\ Disspiated\ During\ Each\ Cycle\ or\ Work\ (W_R)}$$

[1]

State the energy equation associated with the resistor.

During each cycle, the loss resistor will dissipate a time-average (RMS) energy, $W_R = P_R T$

where, energy lost per second, $P_R = \frac{1}{T}\int_0^T p_R(t).dt = \left(\frac{V_R|_{PEAK}}{\sqrt{2}}\right)^2 \frac{1}{R}$ $for\ parallel\ RLC \neq f(t)$

[1]

State the energy equation associated with the capacitor.

$$W_C|_{PEAK} = \frac{1}{2}\ C\ V_C|^2_{PEAK} \quad for\ parallel\ RLC$$

[1]

Using 5(a)(i-iii), state the resulting equation for quality factor.

$$Q(\omega_o) = \omega_o CR \quad for\ parallel\ RLC$$

[1]

ANSWERS

State the frequency associated with 5(a)(iv), in terms of lumped-elements, and define its damping/driven state.

At the undamped (or driven) angular resonance frequency: $\omega_o = \frac{1}{\sqrt{LC}}$

[2]

Model answer to Q5(b): Textbook

The eigenfrequencies for an RLC resonator are given by equation (5.1).

$$\tilde{\omega}_{0,1} = \pm\omega_o\sqrt{1 - \frac{1}{(2Q_u(\omega_o))^2}} + j\frac{\omega_o}{2Q_u(\omega_o)} \qquad (5.1)$$

From equation (5.1), what is the damped (or undriven) natural angular resonance frequency?

$$\omega_o' = \omega_o\sqrt{1 - \frac{1}{(2Q_u(\omega_o))^2}}$$

[1]

From equation (5.1), what is the amplitude decay rate or loss rate or Napier frequency?

$$\omega_o'' = \frac{\omega_o}{2Q_u(\omega_o)} = \frac{1}{\tau''}$$

[1]

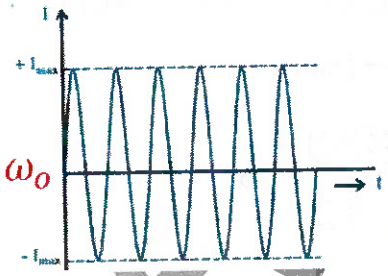
What is the amplitude decay time constant?

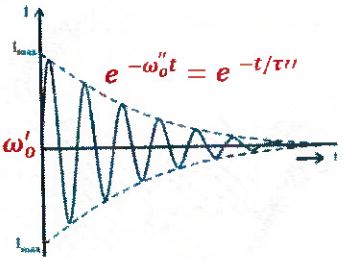
$$\tau'' = \frac{1}{\omega_o''}$$

[1]

Clearly sketch the voltage waveform, as a function of time, for a driven and undriven tank resonator. Show all related variables.

Replace *I* for *V* below!





Field (Amplitude) decay: $e^{st} = e^{j\tilde{\omega}_o t} = e^{j(\omega_o' + j\omega_o'')t} = e^{-\omega_o'' t}e^{+j\omega_o' t} \rightarrow e^{-t/\tau''}sin(\omega_o' t)$

Energy (Power) decay: $e^{2st} =\rightarrow e^{-2t/\tau''}sin(\omega_o' t)$

[2]

Using equation (5.1), derive an expression that relate quality factors at the driven and undriven resonance frequencies.

$$Q_u(\omega_o') = \frac{\omega_o'}{2\omega_o''} \quad \text{where} \quad \omega_o' = \omega_o\sqrt{1 - \frac{1}{(2Q_u(\omega_o))^2}}$$

Manipulating gives: $Q_u(\omega_o') = \sqrt{Q_u(\omega_o)^2 - \frac{1}{4}}$

[2]

T.Swers

With critical damping, what are the values of quality factors at the driven and undriven resonance frequencies?

$$Q_u(\omega_o) = 0.5 \text{ and } Q_u(\omega_o') = 0$$

[1]

Model answer to Q5(c): New Application

i) Undamped resonance frequency.
fo = 866.052 MHz

[1]

ii) Quality factor at the undamped resonance frequency.
Q(fo) = 3.464

[1]

iii) Damped resonance frequency.
fo' = 856.983 MHz

[1]

iv) Quality factor at the damped resonance frequency.
Q(fo') = 3.428

[1]

v) Amplitude decay time constant.

$$\omega_o'' = \frac{\omega_o}{2Q_u(\omega_o)} = \frac{1}{\tau''}$$

τ''= 1.273 ns

[1]

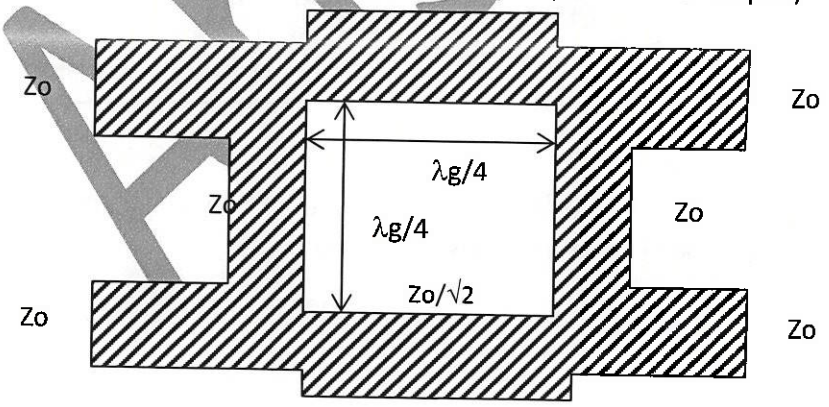
vi) -3 dB bandwidth.
$\Delta f = 2f_o''$ = 250 MHz

[1]

Model answer to Q6(a): New Application

Describe, with the aid of a diagram, a branch-line coupler having a distributed-element implementation, indicating lengths and impedances. Indicate its main characteristics.

90° 3dB Directional Coupler (Branch-line Coupler)



- Works on the interference principle, therefore, narrow fractional bandwidth (15% maximum)
- No bond-wires or isolation resistors required
- Wider tracks make it easier to fabricate and is, therefore, good for lower loss and higher power applications
- Simple design but large
- Meandered lines are possible for lower frequency applications

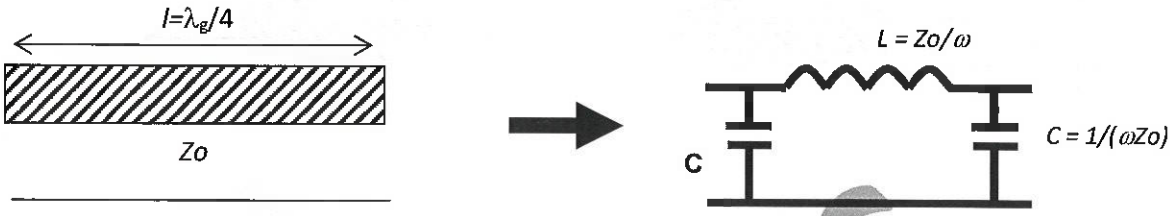
[5]

Answers

Model answer to Q 6(b): Bookwork and Computed Example

Replace the distributed-element implementation in 6(a) with an equivalent lumped-element version. How does the performance compare with that in 6(a)? With a coupler having an impedance of $Zo = 50\ \Omega$, calculate the components values for a design frequency of 1.85 GHz.

The lumped-element equivalent of a $\lambda_g/4$ transmission line is shown below.



All the previous distributed-element couplers can be transformed into equivalent lumped-element couplers by simply replacing all the $\lambda_g/4$ lengths of transmission lines with the above π-network. Since lumped-element components have a lower Q-factor, when compared to distributed-element components, there is an insertion loss penalty. Also, because this π-network is clearly a low-pass filter, having a cut-off frequency, $f_c = \dfrac{1}{2\pi\sqrt{LC}}$, there is also a bandwidth penalty.

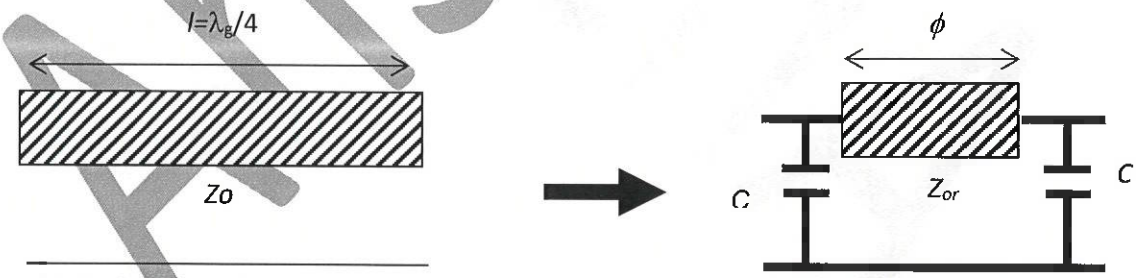
L = 4.3 nH and C = 1.7 pF for the Zo = 50 Ω sections of line
L = 3.0 nH and C = 2.4 pF for the Zo = 35 Ω sections of line

[5]

Model answer to Q 6(c): Bookwork and Computed Example

Replace the lumped-element implementation in 6(b) with an equivalent lumped-distributed version. With a coupler having an impedance of $Zo = 50\ \Omega$ and line sections of electrical length $\phi = 45°$, calculate the components values for a design frequency of 1.85 GHz.

- Lumped-Distributed Couplers



In this 'reduced-size' technique, each $\lambda_g/4$ line is replaced with the above π-network.

$$Z_{or} = \frac{Zo}{Sin\phi} \quad and \quad C = \frac{Cos\phi}{\omega Zo}$$

With $\phi = 45°$,

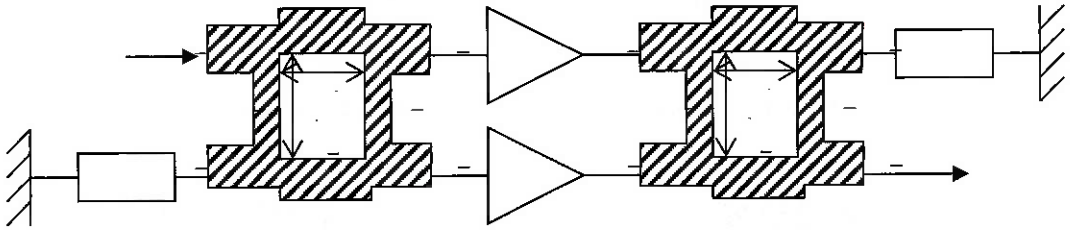
Z_{or} = 70.7 Ω and C = 1.22 pF for the Zo = 50 Ω sections of line
Z_{or} = 50.0 Ω and C = 1.72 pF for the Zo = 35 Ω sections of line

[5]

Answers

Model answer to Q 6(d): Discussion in Class

Draw the topology of a balanced amplifier employing the coupler given in 6(a).



[5]

Answers